

ATOMOD

User guide

H. Bulou

Version 0.6.0

July 7, 2026

Installation

Before you begin, check the ATOMOD systems requirements, to avoid any issues during the process. Ensure you have the following:

- PYTHON:
 - Requires Python 3.12 or an earlier version (e.g., 3.11, 3.10) to avoid compilation errors.
- DISK SPACE:
 - At least 7 GB of free space on your partition.
- SYSTEM SOFTWARE (Required):
 - `git` (to clone the ATOMOD repository)
 - `python3-venv` (to isolate Python packages)
 - `python3-pip` (to install dependencies)
 - `wget` (to download installer `install.py`)

On Ubuntu / Xubuntu / Debian, you can install everything using:

```
sudo apt update && sudo apt install git python3-pip python3-venv wget
```

Windows

Linux

Installation process on linux:

- 1 - Open an X terminal
- 2 - Navigate to the directory of your choice (e.g., `cd ~/Desktop` or `cd ~/src`)(Fig.1(a)).
`cd Desktop`
- 3 - Download the installation script `install.py` (Fig.1(b)).
`wget https://github.com/hbulou/ATOMOD/releases/download/v0.1/install.py`
- 4 - Run the installation (Fig.1(c)). NOTE: You must have at least 7 GB of available disk space.
`python3 install.py`
- 5 - Once the installation is complete, go to the ATOMOD directory (Fig.1(d)).
`cd ATOMOD`

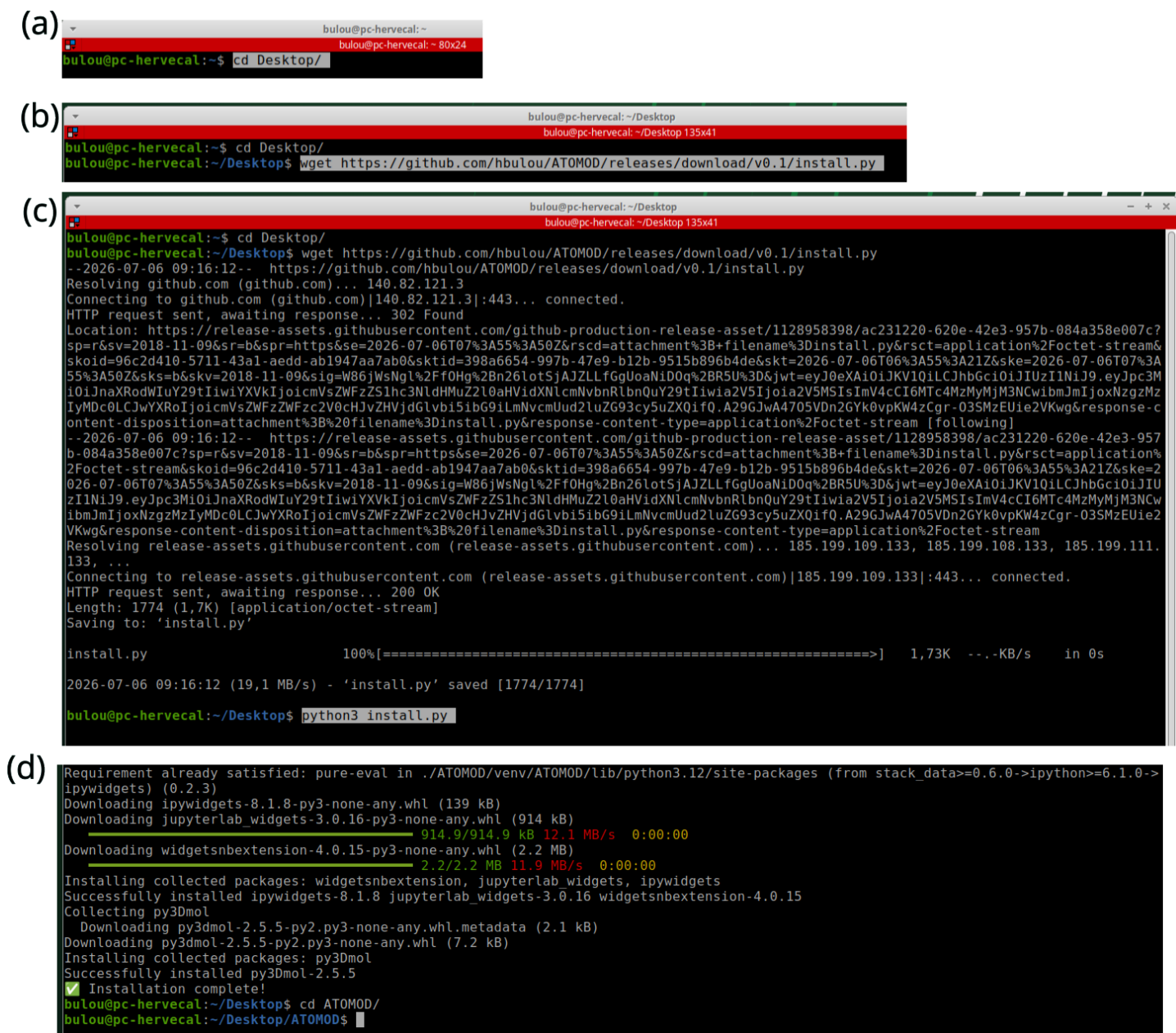


Figure 1: Screenshots of the various stages of installing ATOMOD on Linux.

ATOMOD Update

To update ATOMOS, simply run the `update_ATOMOD.py` script:

```
python3 update_ATOMOD.py
```

Tutorials

Four tutorials are available in the `./doc/tutorials` directory:

- `01-tuto-in-silico-generation.ipynb`, describing the procedure for creating a nanoparticle *in silico*,
- `02-tuto-TEM_image_simulation.ipynb`, describing the procedure for simulating a TEM image,
- `03-tuto-EXAFS.ipynb`, describing the procedure for simulating the EXAFS spectra of a nanoparticle,
- `04-tuto-molecular-dynamics.ipynb`, describing the procedure for performing classical molecular dynamics (MACE force fields).

To run the tutorials, the simplest method is to use the `run_tuto_ATOMOD.py` file located in the `ATOMOD` directory:

```
python3 run_tuto_ATOMOD.py
```